Engineering Mechanics 1

Week 10

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Autonomous Aerial Systems

Principle of Virtual Displacements

- **Work**: Energy transferred to / from an object via the application of a force along a displacement

  Formula: $W = \vec{F} \cdot \vec{s} \Rightarrow \int_{r_1}^{r_2} dW = \int_{r_1}^{r_2} F\,dx$

  $\vec{M} \cdot \delta\vec{\varphi}$

  Constant forces / moments: $\begin{cases} W = \vec{F} \cdot \vec{s} \\ W = \vec{M} \cdot \vec{\varphi} \end{cases}$

- **Virtual displacements**: Shows how the system's trajectory can *hypothetically* deviate from the current status without violating system constraints. Thus, the system needs to be movable.

  Notation: $\delta s_i \rightarrow$ distance. $\delta\varphi_i \rightarrow$ angle

  Characteristics: 1. small displacements

  2. system needs to be movable

  Small An. Assum.

  $\lim_{x \to 0} \sin x = x$

  $\lim_{x \to 0} \cos x = 1$

  $\lim_{x \to 0} \tan x = x$

- **Virtual work**: Total work done by the applied (and inertial) forces of a system as it moves through a set of virtual displacements. If the system is in statics, the virtual work is zero.

  Formula: $\delta W = \sum_{i=2}^{\hat{n}} F_i \delta s_i + \sum_{i=5}^{\hat{n}} M_i \delta\varphi_i$

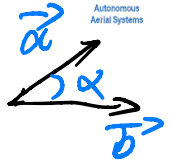
  Statics: $= \sum \vec{F} \cdot d\vec{r} + \sum \vec{M} d\varphi$

  $\delta W = 0$, static: $\sum \vec{F} = 0$, $\sum \vec{M} = 0$.

Principle of Virtual Displacements

- **Examples:**

Dot product : $\vec{a} \cdot \vec{b} = |a||b|\cos\alpha$





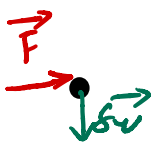
$\delta w = \vec{F} \cdot \delta\vec{u}$



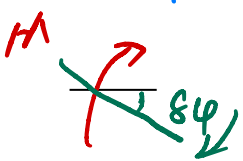
$\delta w = -F \cdot \delta u$



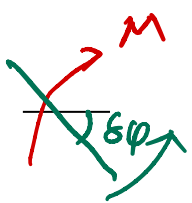
$\delta w = \vec{F} \cdot \delta\vec{u} = |\vec{F}||\delta\vec{u}| \cdot \cos\alpha$

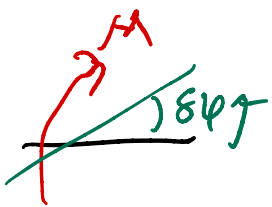


$\delta w = 0$



$\delta w = M \cdot \delta\varphi$



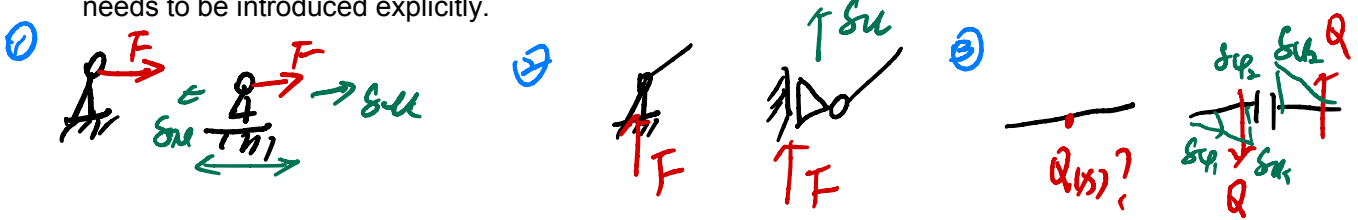


$\delta w = -M \cdot \delta\varphi$

Force calculations with Virtual Work

- **Procedure:**

1. Make system movable by modifying / adding an existing support / joint. The omitted reaction force needs to be introduced explicitly.



2. Construct the pole plane.

3. Draw displacement diagram and get the angle relationships.

A moment/force is applied

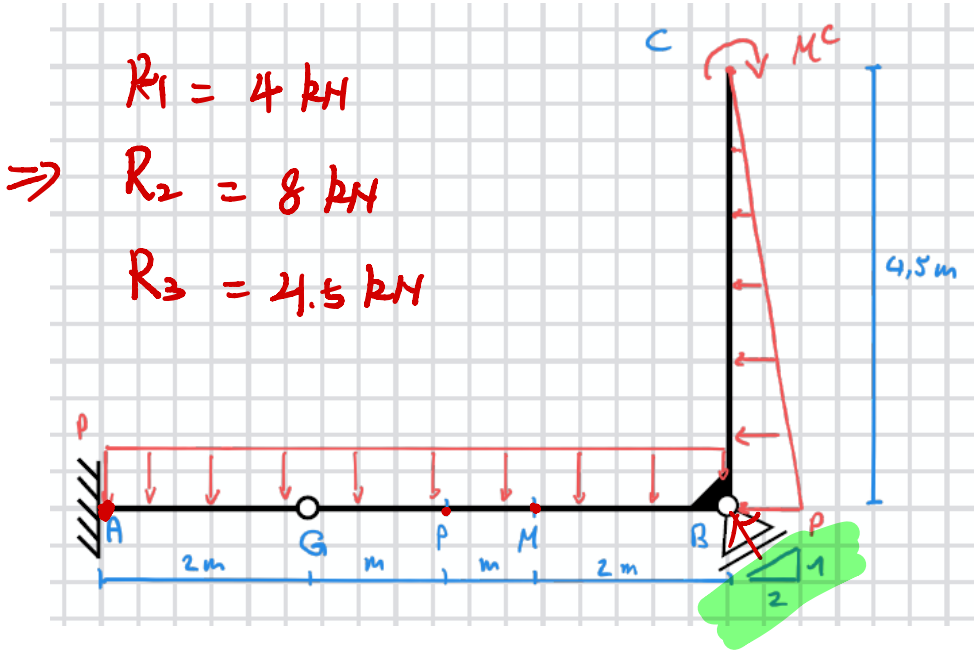
4. Apply the principle of virtual work for statics

Location and values of the displacement.

Our target => Calculate the force or moment in step 1

the only unkown variables or elements

Exercise



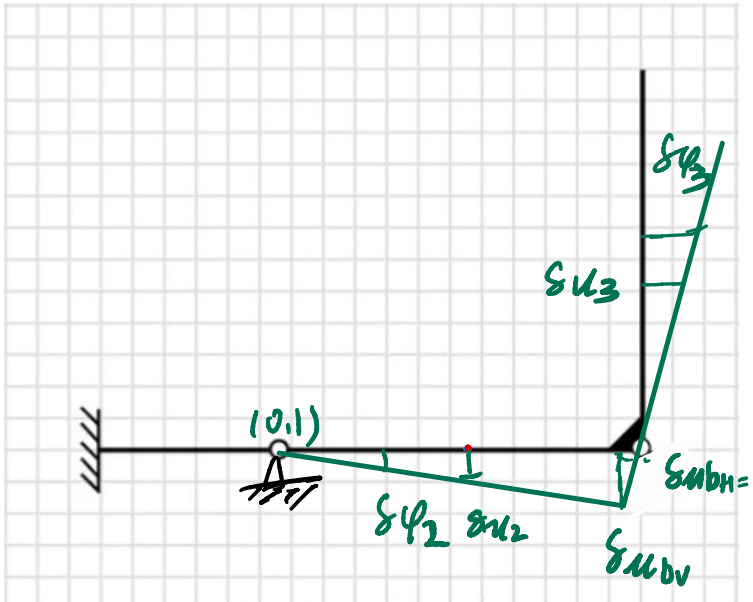
We know

- p = 2 kN/m
- M^c = 0.75 kNm

Calculate

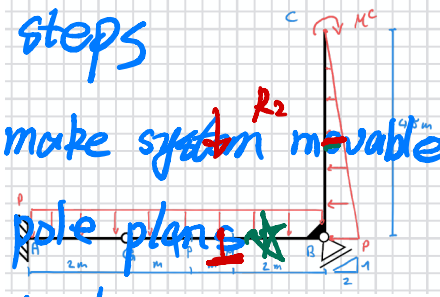
- Support reactions: F^b, M^a
- Internal forces: M^m,Q^p

** with principle of virtual work



4 steps

1. make system movable



2. pole plan

3. displacement diagram

4. Apply δW

F^b

$$\delta u_2 = \frac{1}{2} l_{R_2} \cdot \delta\varphi_2 \quad ①$$

$$\delta u_3 = \frac{1}{3} l_{R_3} \cdot \delta\varphi_3 \quad ②$$

$$\delta u_{bv} = l_{R_2} \delta\varphi_1 \quad ③. \quad \delta u_{bH} = 0.$$

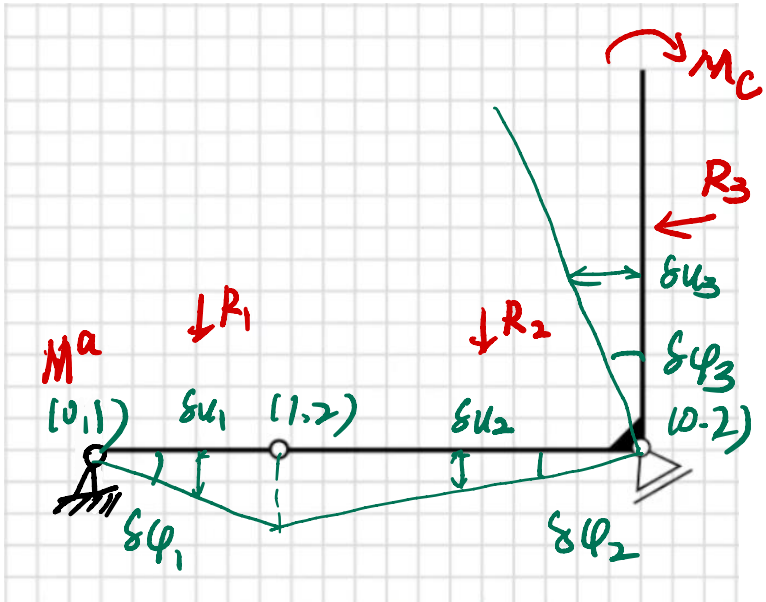
Virtual work

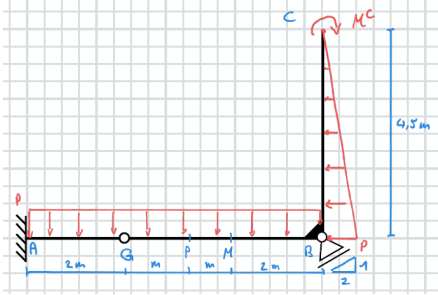
$$\delta w = 0$$

$$= -F_V^b \cdot \delta u_{bv} + R_2 \cdot \delta u_2 - R_3 \cdot \delta u_3 + M_c \delta\varphi_3 = 0 \quad ④$$

$$\Rightarrow \quad F^b = \frac{N\bar{5}}{2} F_V^b \doteq 2.8\ kN.$$

M^a





4 steps

1. make system movable
2. pole plans ★
3. displacement diagram.
4. Apply $\delta w = 0$

$$\delta u_1 = \frac{1}{2} L_{R_1} \cdot \delta\varphi_1 = \delta\varphi_1 \quad ①$$

$$\delta u_2 = \frac{1}{2} L_{R_2} \cdot \delta\varphi_2 = 2\delta\varphi_2 \quad ②$$

$$\delta u_3 = \frac{1}{3} L_{R_3} \cdot \delta\varphi_3 = 1.5\delta\varphi_3 \quad ③$$

$$\delta\varphi_1 = 2\delta\varphi_2$$

$$\delta\varphi_3 = \delta\varphi_2$$

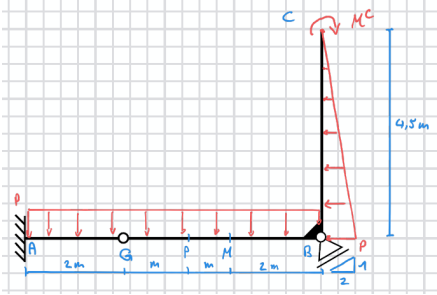
$$\delta W = 0$$

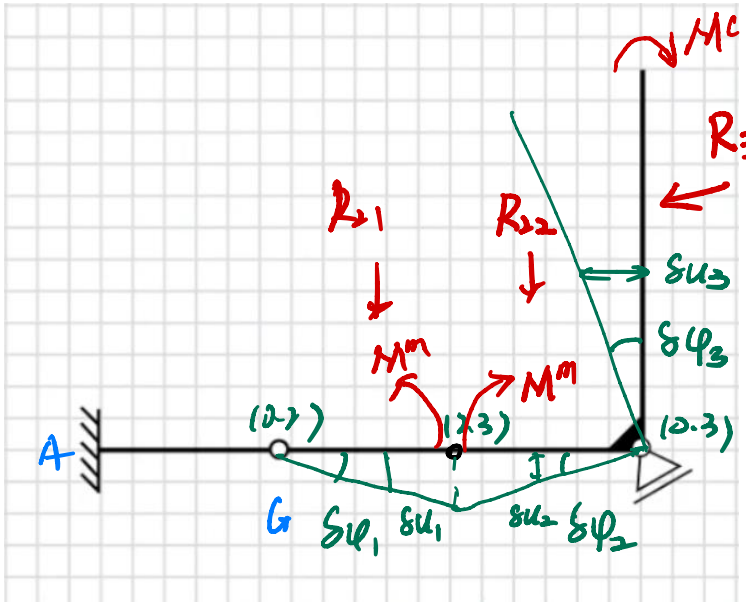
$$= M^a \cdot \delta\varphi_1 + R_1\delta u_1 + R_2\delta u_2 + R_3\delta u_3 - M_c\delta\varphi_3 = 0 \quad ④$$

$$= \delta\varphi_2 (2M^a + 2R_1 + 2R_2 + 1.5R_3 - M_c) = 0$$

$$M^a = \frac{M_c - 1.5R_3 - 2(R_1 + R_2)}{2} = -15\ \text{kN·m}$$

M^m





4 steps

1. make system movable
2. pole plans ☆
3. displacement diagram ☆
4. Apply δw.

$$\delta u_1 = \frac{1}{2} L_{R_{21}} \cdot \delta\varphi_1 \quad ①$$

$$\delta u_2 = \frac{1}{2} L_{R_{22}} \cdot \delta\varphi_2 \quad ②$$

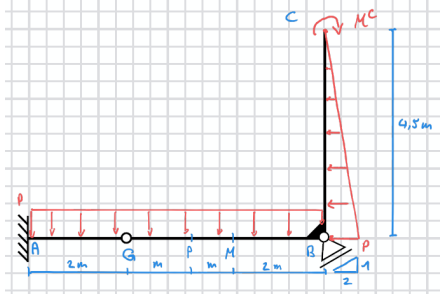
$$\delta u_3 = \frac{1}{3} L_{R_3} \cdot \delta\varphi_3 \quad ③$$

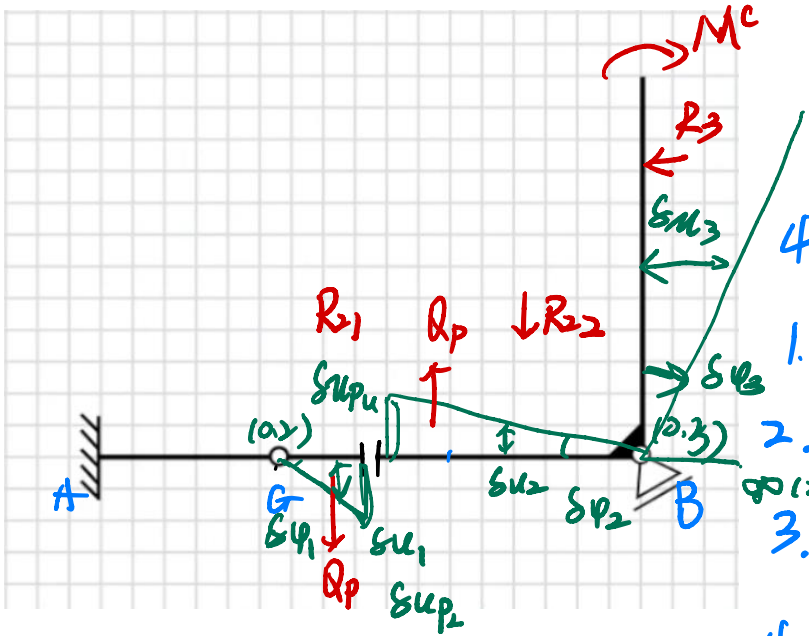
$$\delta\varphi_1 = \delta\varphi_2 = \delta\varphi_3$$

$$\delta w = 0$$

$$= -2M^m \cdot \delta\varphi_1 + R_{21} \cdot \delta u_1 + R_{22} \cdot \delta u_2 + R_3 \delta u_3 - M_c \delta\varphi_3 = 0$$

$$M^m = \frac{R_{21} + R_{22} + 1.5R_3 - M_c}{2} = 7 \text{ kN.m}$$





4 steps

1. make system movable
2. pole plans ☆
3. displacement diagram ☆
4. Apply δw

$$\delta u_1 = \frac{1}{2} L_{R_{21}} \cdot \delta\varphi_1 \quad ①$$

$$\delta u_2 = \frac{1}{2} L_{R_{22}} \cdot \delta\varphi_2 \quad ②$$

$$\delta u_3 = \frac{1}{3} L_{R_3} \cdot \delta\varphi_2 \quad ③$$

$$\delta u_{P_V} = L_{R_{22}} \cdot \delta\varphi_2$$

$$\delta u_{P_2} = L_{R_{21}} \cdot \delta\varphi_1$$

$$\delta\varphi_1 = 3\delta\varphi_2$$

$$\delta w = 0$$

$$= Q^P \delta u_{P_V} + Q^P \delta u_{P_2} + R_{21} \cdot \delta u_1 - R_{22} \cdot \delta u_2 - R_3 \cdot \delta u_3 + M^C \cdot \delta\varphi_3$$

$$Q^P = \frac{-0.5R_{21} + 1.5R_{22} + 1.5R_3 - M_C}{4} = 3.5\ kN$$